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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

df = pd.read_csv('truck.csv')
df.head()
```

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Out[1]:
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	X	Y
0	12.4	11.2
1	14.3	12.5
2	14.5	12.7
3	14.9	13.1
4	16.1	14.1

```
In [2]: r, p_value = stats.pearsonr(df['X'], df['Y'])
print(f"Pearson's r = {r:.4f}")
print(f"p-value = {p_value:.3e}")
```

Pearson's r = 0.9985
p-value = 4.148e-28

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In [3]: slope, intercept, r_value, p_value_lr, std_err = stats.linregress(df['X'], df['Y'])
print(f"Best Price = {slope:.4f} * List Price + {intercept:.4f}")
print(f"R-squared = {r_value**2:.4f}")
```

Best Price = 0.8511 * List Price + 0.4346
R-squared = 0.9971

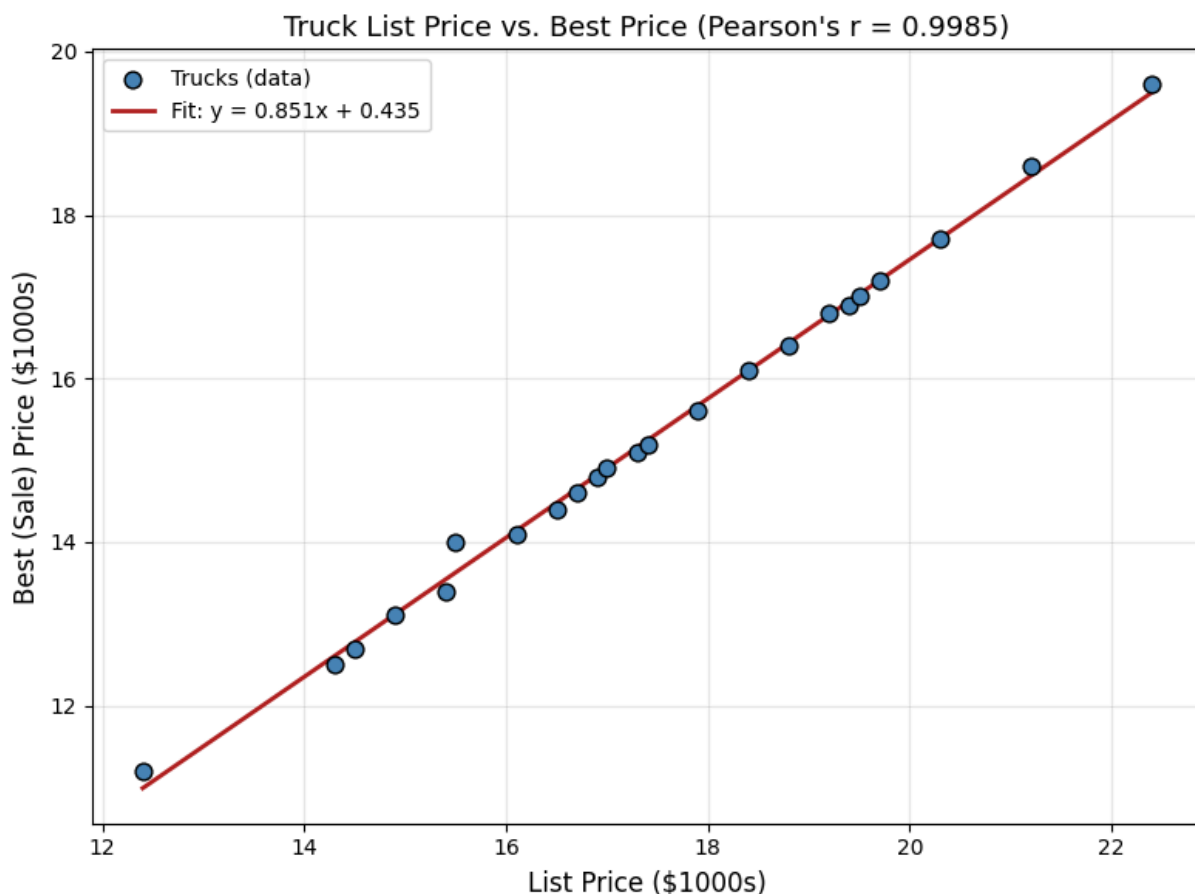
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In [4]: fig, ax = plt.subplots(figsize=(8, 6))

ax.scatter(df['X'], df['Y'], color='steelblue', edgecolor='black', s=60, label='Data')

x_line = np.linspace(df['X'].min(), df['X'].max(), 100)
y_line = slope * x_line + intercept
ax.plot(x_line, y_line, color='firebrick', linewidth=2, label=f'Fit: y = {slope:.4f}x + {intercept:.4f}')

ax.set_xlabel('List Price ($1000s)', fontsize=12)
ax.set_ylabel('Best (Sale) Price ($1000s)', fontsize=12)
ax.set_title(f'Truck List Price vs. Best Price (Pearson's r = {r:.4f})", for
ax.legend()
ax.grid(alpha=0.3)

plt.tight_layout()
plt.savefig('truck_scatter.png', dpi=150)
plt.show()
```



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In [5]: list_price_choice = 20.0
predicted_best_price = slope * list_price_choice + intercept
print(f"Predicted best price for a list price of ${list_price_choice*1000:,.0f},.
      f"${predicted_best_price*1000:,.0f}")
```

Predicted best price for a list price of \$20,000: \$17,457

The correlation is very strong ($r = 0.9985$ and $R^2 = 0.9971$), so the linear relationship is very strong. This is very visible in the graph. My prediction falls on the line. I am very confident because of this.